Sequences, Series, Limits & Continuity

A *sequence* of real numbers is a function, u, mapping $\mathbb{N}$ to $\mathbb{R}$

- We will be dealing with infinite sequences
- Written $u(1), u(2), u(3), ...$, or $(u_1, u_2, u_3, ...)$
- The sequence is written $\{u_n\}$
- Example of a sequence

$$u_n = \frac{3n^2}{n^3 + 9}$$

which gives the sequence

$$\frac{3}{10}, \frac{12}{17}, \frac{48}{73}, ...$$

Arithmetic sequences

- An arithmetic sequence is a sequence $\{u_n\}$ in which

$$u_{n+1} - u_n$$

  is constant for all n

- Equivalent to saying that for some numbers a, d, the sequence looks like

$$a, a + d, a + 2d, a + 3d, ...$$

  or

$$u_n = a + (n - 1)d$$

- Is this sequence arithmetic? (If so, what are a, d?)

$$u_n = 14 - 5n$$

- Is this one?

$$u_n = 5^{-n}$$

Arithmetic sequences: Each term is obtained by adding same number to previous term

Geometric sequences: Each term is obtained by multiplying previous term by same number

- Equivalent to saying that for some numbers a, x the sequence looks like

$$a, ax, ax^2, ax^3, \ldots$$

or

$$u_n = ax^{n-1}$$

- What are a, x for this geometric sequence?

$$u_n = 5^{-n}$$

Limits of sequences

- The *limit* of a sequence describes what happens to u_n as n becomes large
- A sequence $\{u_n\}$ tends to the limit z if for *any* positive real number ϵ there is a positive integer, N, such that:

$$|u_n - z| < \epsilon \text{ for all } n > N.$$

- Also written as

$$\lim u_n = z$$

$$\lim_{n \to \infty} u_n = z$$

$$u_n \to z \text{ as } n \to \infty$$

An example: Let's show that $u_n = \frac{1}{n}$ tends to limit $z = 0$

Convergent sequence: $\lim u_n = z$ if for *any* $\epsilon > 0$ there is a positive integer, N such that $|u_n - z| < \epsilon$ for all $n > N$.

Put differently, for any $\epsilon > 0$, **IF** $n > N$ **THEN** $|u_n - z| < \epsilon$.

Take any $\epsilon > 0$ (it can be arbitrarily small). Let N be any integer greater than $\frac{1}{\epsilon}$. Note that $N > \frac{1}{\epsilon}$ implies that $\epsilon > \frac{1}{N}$.

Then for all $n > N$, we have

$$|u_n - 0| = \frac{1}{n} < \frac{1}{N} < \epsilon.$$

It follows that our sequence $u_n = \frac{1}{n}$ tends to the limit $z = 0$.

Definitions:

A sequence that tends to a limit is called *convergent*; a sequence that doesn't is called *divergent*

A sequence that tends to infinity (or negative infinity) is divergent, but a sequence doesn't have to tend to infinity to be divergent (what's an example of a divergent sequence that doesn't tend to infinity?)

A *Cauchy* sequence of real numbers is a sequence whose elements cluster around some number as n gets large. Since we are dealing with the real numbers, a Cauchy sequence is equivalent to a convergent sequence.

$$\{x_n\} \text{ is Cauchy if } \forall \epsilon > 0, \exists N \in \mathbb{N} : |x_m - x_n| < \epsilon \text{ when } m, n > N$$

$\{x_n\}$ is Cauchy if $\forall \epsilon > 0, \exists N \in \mathbb{N} : |x_m - x_n| < \epsilon$ when $m, n > N$

We can show the sequence defined by $x_n = n$ is not convergent:

$|x_n - x_m| = |n - m| \geq 1$ for all n, m. Thus, for $\epsilon = 1$ there is no N such that $|x_n - x_m| < \epsilon$ whenever $n, m > N$. Therefore $\{x_n\}$ is not Cauchy, and does not tend to a limit.

Working with limits

Fact 1: For a real number x, $\lim_{n\to\infty} x^n = 0$ if and only if $-1 < x < 1$.

Fact 2: For a real number b, $\lim_{n\to\infty} n^b = 0$ if and only if $b < 0$.

As $n \to \infty$, what's the limit of

$$u_n = \frac{n+1}{2n}$$

Claim: If $\lim u_n = z$ then $\lim |u_n| = |z|$

Proof: If $\lim u_n = z$ then for any $\epsilon > 0$, $|u_n - z| < \epsilon$ for n large. First, note that for real numbers a, b we have

$$||a| - |b|| \leq |a - b|$$

(the "reverse triangle inequality"). This basically proves our result: For any $\epsilon > 0$ we have $|u_n - z| < \epsilon$ for n large, and consequently $||u_n| - |z|| < |u_n - z| < \epsilon$. Thus, $\lim |u_n| = |z|$. $\square$

Does the converse hold? (If $\lim |u_n| = |z|$ then $\lim u_n = z$)

What is the limit of

$$u_n = \frac{(-1)^n (n+1)^2}{n^4}?$$

Hint: We can start by expanding the numerator.

What would happen if we replaced $(n+1)^2$ with $(n+1)^4$?
Don't do whole calculation!

Series

- The series defined by a sequence $\{u_n\}$ is the sequence $\{s_n\}$ with

$$s_1 = u_1, s_2 = u_1 + u_2, s_3 = u_1 + u_2 + u_3, ...$$

and more generally

$$s_n = \sum_{r=1}^{n} u_r$$

Summing algebraic sequences

First, the natural numbers

$$s_n = 1 + 2 + ... + (n - 1) + n$$

$$s_n = n + (n - 1) + ... + 2 + 1$$

Adding these together we get

$$2s_n = (1 + n) + (1 + n) + ... + (1 + n) = n(1 + n)$$

The first n terms of the natural numbers sum to

$$s_n = \frac{1}{2}n(1 + n)$$

Recall that a general arithmetic sequence is $u_n = a + (n-1)d$

We can use the same trick to find the sum of the first n terms of the sequence:

$$u_n = a + (a+d) + (a+2d) + ... + (a+(n-2)d) + (a+(n-1)d)$$

$$u_n = (a+(n-1)d) + (a+(n-2)d) + ... + (a+2d) + (a+d) + a$$

$$2u_n = n\,(2a + (n-1)d)$$

$$u_n = an + n\left(\frac{(n-1)d}{2}\right)$$

Summing geometric sequences: $s_n = a + ax + ax^2 + ... + ax^{n-1}$

Multiplying by x we get

$$xs_n = ax + ax^2 + ... + ax^n$$

Let

$$t = ax + ax^2 + ... + ax^{n-1}$$

Then $s_n = a + t$ and $xs_n = t + ax^n$

$$s_n - xs_n = a - ax^n$$

so

$$s_n(1 - x) = a - ax^n,$$

and finally

$$s_n = a\left(\frac{1 - x^n}{1 - x}\right)$$

Convergence of geometric series

We will often care about the limit of a series as $n \to \infty$

For a geometric series,

$$s_n = a\left(\frac{1-x^n}{1-x}\right)$$

We know that $x^n \to 0$ as $n \to \infty$ if and only if $|x| < 1$

If $|x| < 1$ then $\lim_{n\to\infty} s_n = a\left(\frac{1-0}{1-x}\right) = \frac{a}{1-x}$

Divergence of geometric series

Fact (*term test* for any series): If $\lim_{n\to\infty} k_n \neq 0$ then $\sum_{n=1}^{\infty} k_n$ diverges

For a geometric series, $k_n = ax^{n-1}$

We know that $x^n \to 0$ as $n \to \infty$ <u>if and only if</u> $|x| < 1$

Therefore, a geometric series diverges whenever $|x| \geq 1$

Ex: Sequences in the social sciences (*exponential discounting*)

Suppose that I plan to consume x units of cake in each of an infinite number of time periods, $t = 1, 2, 3, ...$

I discount the utility I'll receive from future consumption by a term $0 < \delta < 1$, so that I value today's consumption fully as $u(x)$, tomorrow's as $\delta u(x)$, the day after's as $\delta^2 u(x)$...

My expected utility over time is

$$\sum_{t=0}^{\infty} \delta^t u(x) = \frac{u(x)}{1-\delta}$$

"Cake eating problem": How to allocate a fixed unit of x over this infinite stream *today* in order to maximize my lifetime utility

Limits and continuity

Let $f(x) = y$ be a real-valued function, where the domain of f is either all the real numbers or some interval of real numbers

The *domain* of $f : X \to Y$ is X, the "input" values of f;
Y is the *codomain*

The *range* of f (or *image*) is the set of values of Y that f actually takes

$f(x)$ "tends to the limit l as x approaches x_o"

$$f(x) \to l \text{ as } x \to x_o$$

$$\lim_{x \to x_o} f(x) = l$$

These statements mean that $f(x)$ can attain values as close as we want to l for all x sufficiently close (but not equal) to x_o

More formally:

$$f(x) \to l \text{ as } x \to x_o$$

if for any $\epsilon > 0$ we can choose a $\delta > 0$ such that

$$|f(x) - l| < \epsilon \text{ for all } x \text{ such that } 0 < |x - x_o| < \delta$$

Example: $f(x) = 20 - 10x$. Then $\lim_{x \to 1} f(x) = 10$. To prove this using (δ, ϵ) we want to show that for *any* $\epsilon > 0$ we can find a $\delta > 0$ so that

$$|f(x) - 10| < \epsilon \text{ whenever } |x - 1| < \delta$$

$$|f(x) - 10| < \epsilon \text{ whenever } |x - 1| < \delta$$

$$|(20 - 10x) - 10| < \epsilon \text{ whenever } |x - 1| < \delta$$

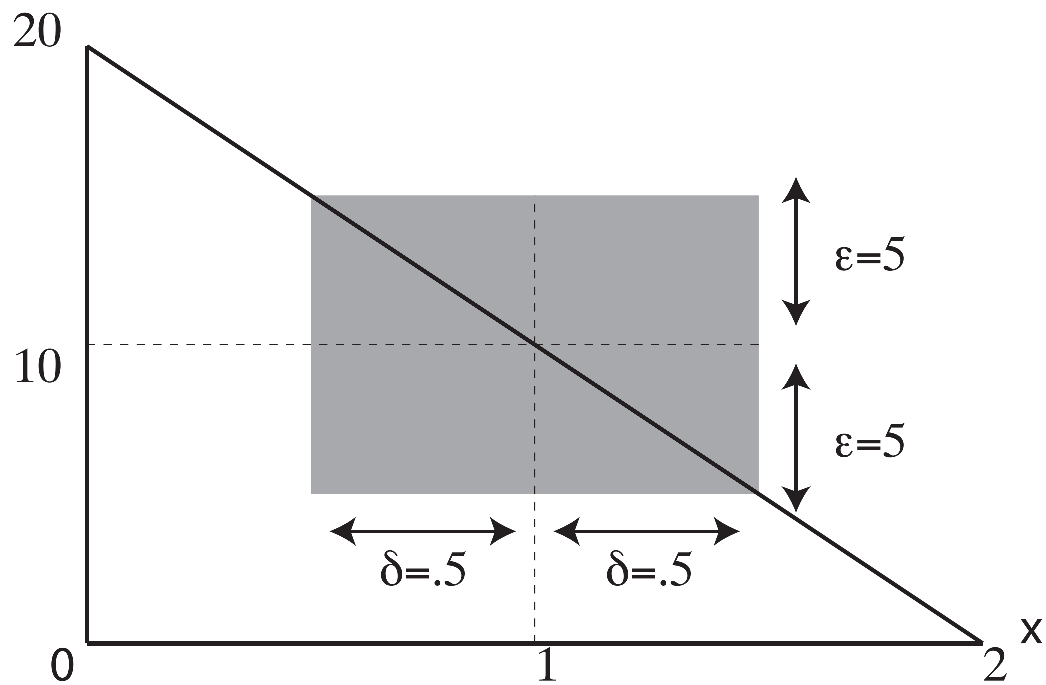
$$|10 - 10x| < \epsilon \text{ whenever } |x - 1| < \delta$$

$$10|x - 1| < \epsilon \text{ whenever } |x - 1| < \frac{\epsilon}{10}$$

For any ϵ we could choose, this holds for $\delta = \frac{\epsilon}{10}$
(Plug in some examples: try $\epsilon = 5$)

$f(x) = 20 - 10x$

$\lim_{x \to 1} f(x) = 10$



Important points:

$\lim_{x \to x_o} f(x)$ might or might not equal $f(x_o)$

- It could be *different than* $f(x_o)$

$$f(x) = \begin{cases} +1 & \text{if } x \neq 4 \\ -1 & \text{if } x = 4 \end{cases}$$

- It could be *undefined*

$$f(x) = \begin{cases} +1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

Operations with limits

Suppose that $f(x) \to l$ and $g(x) \to m$ as $x \to x_o$. Then:

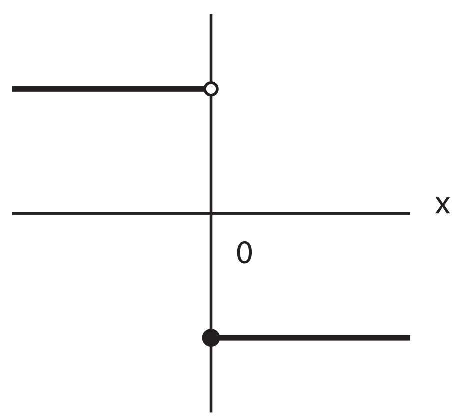
1. $\lim_{x \to x_o} f(x) + g(x) = l + m$
2. $\lim_{x \to x_o} f(x)g(x) = lm$
3. $\lim_{x \to x_o} \frac{f(x)}{g(x)} = \frac{l}{m}$ if $m \neq 0$
   - For this last case, if $m, l = 0$ or $m, l = \infty$ we have an *indeterminate form*, and the limit of $\frac{f(x)}{g(x)}$ might or might not exist. We use L'Hôpital's Rule in this case (requires calculus so we'll hold off on this).

One-sided limits

$f(x)$ tends to l as x approaches x_o from above (below):
$\lim_{x \downarrow x_o} f(x) = l$

$$f(x) = \begin{cases} +1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

In this case, $\lim_{x \downarrow 0} f(x) = -1$ and $\lim_{x \uparrow 0} f(x) = 1$



Continuity

- If $\lim_{x \to x_o} f(x) = f(x_o)$ we say that f is *continuous* at x_o; if not, the function is *discontinuous* at x_o
- If a function f is continuous for every x then f is a *continuous function*
- Most important thing about continuous functions: f is continuous if arbitrarily small changes in x produce arbitrarily small changes in $f(x)$. These changes can be as small as we want them to be.
- In other words, we can make the change in $f(x)$ as small as we like if we make the change in x small enough.

Continuity: Small changes in x map into small changes in $f(x)$

A function $f : \mathbb{R} \to \mathbb{R}$ is continuous at point $x_o \in \mathbb{R}$ if for any $\epsilon > 0$ there exists a $\delta > 0$ such that

$$\text{If } |x_o - x| < \delta \text{ then} |f(x_o) - f(x)| < \epsilon$$

The intermediate value theorem

If the function $f(x) = y$ is continuous for all $a \leq x \leq b$ then f takes every value between $f(a)$ and $f(b)$